

the science of ambient electronics is all about.

Something like home décor?

Whether to call ambient electronics a science, an art or a vision is a matter of subjectivity. Whether we have real ambient electronics around us is also a matter of opinion.

Ambient electronics is nothing but the devices that are all around you as part of your ambience or environment. There are devices all around you even today—computer, modem, mobile, Internet-enabled television and more. But you see them as devices and are conscious of their presence.

In the ambient electronics era, all these devices will blend with your home décor. Your oven, washing machine, water heater and door, everything will have a computer embedded in it. Almost every object in your house will be equipped with sensors, embedded systems, communications, displays and input capability. But these devices will be inconspicuous, silent helpers.

“Ambient electronics is essentially electronics (gadgets) becoming a seamless, inalienable part of the ecosystem. It can be smart devices, provided these devices transcend to being integral to and an indistinguishable part of everyday life. In that sense, it is not just ‘smartness’ in functionality but appearing smart as well,” explains S. Uma Mahesh, founder-CEO of Indrion Technologies—one of the key Indian start-ups working in this space.

When ambient electronics is mainstream and mature enough to be integrated, the end result is likely to be electronic environments that are sensitive and responsive to the presence of people. Eli Zelkha and Brian Epstein of Palo Alto Ventures termed this stage as ambient intelligence and predicted it to materialise around 2020.

Wikipedia explains: “In an ambient intelligence world, devices work in concert to support people in carrying out their everyday activities, tasks and



Home automation keypads

rituals in an easy, natural way using information and intelligence that is hidden in the network connecting these devices (the Internet of Things). As these devices grow smaller, more connected and more integrated into our environment, the technology disappears into our surroundings until only the user interface remains perceivable by users.”

Building the future, one ambient device at a time

The moment we think of “A computer in every object,” lots of keywords come to our mind—the Internet of Things, pervasive computing, ubiquitous computing, smart home and so on—for the idea has been spoken about so much. The next question that arises is “How is ambient electronics different from or related to these?”

Chiaki Ishikawa of the YRP Ubiquitous Networking Laboratory, Tokyo, Japan, clarifies that ambient electronics actually forms the foundation of smart or ubiquitous computing that sits on top of them. Add a bit of distributed or centralised control to ambient electronics, and you have a more connected, and collectively intelligent system, which might then be called ubiquitous computing, ambient intelligence, calm computing or whatever.

The future is likely to be an amal-

gamation of many concepts like pervasive computing, ubiquitous computing, profiling practices and human-centric computer interaction design. But the prerequisite for all this is the availability of truly ambient devices.

While some ambient devices are already available today, implementing all the concepts of ambient electronics and intelligence would take many more years. This is because the design and deployment of ambient electronics poses unique challenges, as is evident from the name itself. In short, these devices have to blend with the ambience—which means they need to be small, flexible, capable of being weaved into other objects, and aesthetic too.

A calm entrance; and golden rules

Ambient intelligence—a concept which emerged more than 20 years ago—might take 20 more years to materialise.

Dr Mark Weiser of Xerox PARC termed the concept of ambient electronics and ubiquitous computing as the third era in computing, which he believed had already begun when he wrote about it in 1988. The first two eras of mainframes and personal computers were dominated by machines. Dr Weiser foresaw a third wave when